TEXORA · LECTURE SHEET

Factorial Study of Seam Strength

Design of experiments on the sewing floor: a complete 2^2 factorial study of seam strength — stitch density × seam allowance — with Yates-order effects, ANOVA, the interaction plot, and the engineering conclusion drawn correctly from all three.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 7 · LEVEL 4 TERM 2

1. Why factorial, and why seam strength

Seam strength decides whether a garment survives wear: it depends on stitch density, seam allowance, thread, needle, fabric and their combinations. The traditional way to study it — change **one factor at a time** (OFAT) — is doubly defective: it needs many runs, and it is structurally blind to **interactions** (cases where the effect of one factor depends on the level of another — exactly what happens when extra stitches only help if the allowance is wide enough to hold them). A **factorial design** runs every combination of factor levels, so each effect is estimated from all the data and interactions are measured, not missed. This sheet runs the smallest useful case completely: two factors at two levels (2^2), three replicates per cell, analysed exactly as an IE/quality engineer would report it (linking to Topic 8).

2. The design

FACTOR	LOW LEVEL (−)	HIGH LEVEL (+)	HELD CONSTANT
A — stitch density	4 stitches/cm	6 stitches/cm	Fabric (one lot, warp direction), seam type SSa (TE-3108 Study 1), stitch 301, thread tex 24 core-spun, needle Nm 80, operator, machine, test: grab seam-strength method, 3 specimens per cell.
B — seam allowance	1.0 cm	1.5 cm	

Runs in standard (Yates) order, with the replicate results and cell means (seam breaking force, N):

RUN	A	B	REPLICATES (N)	MEAN (N)	TOTAL
(1)	−	−	372, 385, 383	380	1,140

a	+	−	415, 428, 417	420	1,260
b	−	+	392, 406, 402	400	1,200
ab	+	+	462, 478, 470	470	1,410

Run order in the laboratory was randomised (not the table order) so that drift in machine or operator cannot masquerade as a factor effect; replicates give the pure-error estimate the ANOVA needs. Grand mean = (380 + 420 + 400 + 470) ÷ 4 = **417.5 N**.

3. Effects: the Yates calculation

EFFECTS FROM THE CELL MEANS — VERIFY EVERY NUMBER

An effect is the average change in response when the factor moves from − to +, averaged over the other factor's levels.

Effect A (stitch density) = [−(1) + a − b + ab] ÷ 2 = [−380 + 420 − 400 + 470] ÷ 2 = 110 ÷ 2 = **+55 N**.

Effect B (allowance) = [−(1) − a + b + ab] ÷ 2 = [−380 − 420 + 400 + 470] ÷ 2 = 70 ÷ 2 = **+35 N**.

Effect AB (interaction) = [(1) − a − b + ab] ÷ 2 = [380 − 420 − 400 + 470] ÷ 2 = 30 ÷ 2 = **+15 N**.

Reading: raising stitch density adds on average 55 N; widening the allowance adds 35 N; and the two help each other by a further 15 N of interaction — at wide allowance the density effect is 470 − 400 = 70 N, at narrow allowance only 420 − 380 = 40 N. The fitted value at ab: 417.5 + 55/2 + 35/2 + 15/2 = **470 N**, reproducing the cell mean exactly (the 2^2 model is saturated).

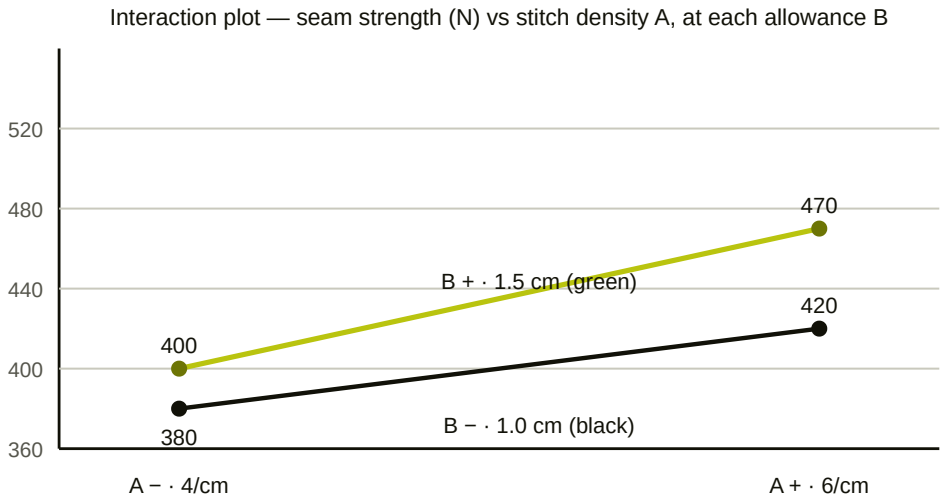


Figure 1 — Interaction plot from the cell means (plot scale: 360 N at y = 230, 40 N per 40 px). The lines are not parallel — the density gain is 40 N at B− and 70 N at B+ — which is the +15 N interaction made visible.

4. ANOVA: are the effects real?

Sums of squares use the totals: SS = contrast² ÷ (n × 2²) with n = 3 replicates. Contrasts on totals: A: −1,140 + 1,260 − 1,200 + 1,410 = 330 → SS_A = 330² ÷ 12 = **9,075**. B: contrast 210 → SS_B = 210² ÷ 12 = **3,675**. AB: contrast 90 → SS_{AB} = 90² ÷ 12 = **675**. Pure error from replicates: SS_E = **428** on 8 degrees of freedom → MS_E = 53.5. (Check: 9,075 + 3,675 + 675 + 428 = 13,853 = total SS.)

SOURCE	SS	DF	MS	F = MS ÷ 53.5	% OF TOTAL
A — stitch density	9,075	1	9,075	169.6	65.5%
B — seam allowance	3,675	1	3,675	68.7	26.5%
AB — interaction	675	1	675	12.6	4.9%
Error	428	8	53.5	—	3.1%
Total	13,853	11			100%

The 5% critical value F(1, 8) = 5.32. All three F ratios exceed it comfortably: **both main effects and the interaction are statistically significant**. But significance is not importance: the percentage column is the engineering story — stitch density carries two-thirds of the explained variation, allowance a quarter, interaction a twentieth. Spend effort in that order.

5. Engineering conclusions

- **Best setting in the study:** A+ B+ (6 stitches/cm, 1.5 cm allowance) at 470 N mean. Against the base fabric strength of 520 N (same grab method), seam efficiency = 470 ÷ 520 × 100 = **90.4%** — a strong seam; at (1) it is only 380 ÷ 520 = 73.1%.
- **Costs of the settings:** density costs thread and sewing time (and beyond a limit, needle damage and puckering — outside this design's range, a warning against extrapolating the line beyond 6/cm); allowance costs fabric at every seam (Topic 5's consumption arithmetic prices it). The factorial result prices the benefit side of both trades.
- **Interaction discipline:** an OFAT study at B− would have reported the density effect as 40 N and undervalued it for the wide-allowance products where it is worth 70 N. That is the standing argument for factorial designs in process work.
- **Next design:** add centre points or a third level to test curvature (is 7/cm better still, or does needle damage reverse the trend?), and extend factors to thread size in a 2^3 — the same machinery scales.

Glossary

TERM	MEANING
Factor / level	A variable deliberately changed in the experiment / one of its set values.
Main effect	Average response change when one factor moves low → high, over all levels of the others.
Interaction	The part of the response change not explained by main effects — one factor's effect depending on another's level.
Replicate	An independent repeat of a treatment combination; provides the pure-error estimate.
Seam efficiency	Seam strength ÷ fabric strength × 100, same test method.

Key points and summary

- Factorial designs estimate all main effects and interactions from every run; OFAT cannot see interactions.
- Worked study: A (density) +55 N, B (allowance) +35 N, AB +15 N; all significant ($F > 5.32$); A dominates at 65.5% of variation.
- Best cell ab: 470 N, seam efficiency 90.4% — adopt with its thread/fabric costs priced.
- Randomise run order, replicate for error, and never extrapolate a two-level design beyond its range.

Practice

Answer in your own words; brief model answers follow.

1. Using the cell means, compute the effect of B at A− and at A+ separately, and show their average is the main effect of B and their half-difference is the AB interaction.
2. What F ratio would AB need to exceed to be significant at 5% here, and why is AB still the least important term despite passing?
3. A manager proposes testing density at 1.0 cm allowance only, "to save specimens". State what the conclusion would have been and why it misleads for 1.5 cm products.
4. Calculate seam efficiency at run b (400 N) against the 520 N fabric.
5. Why was run order randomised when the analysis uses only cell totals?

Answer key

1. At A−: 400 − 380 = +20 N. At A+: 470 − 420 = +50 N. Average = (20 + 50) ÷ 2 = 35 N = main effect B. Half-difference = (50 − 20) ÷ 2 = 15 N = AB interaction. (Symmetrically for A: 40 and 70

N.)

2. F > 5.32 (the 5% point of F(1,8)). It passes — it is real — but it explains only 4.9% of total variation, so process-improvement effort belongs first to A and B; statistical significance and engineering importance are different judgements.
3. Density effect at B− = 420 − 380 = 40 N — reported as "the" density effect. For 1.5 cm products the true effect is 70 N; specifications and thread budgets set from 40 N would under-engineer exactly the products that gain most.
4. 400 ÷ 520 × 100 = 76.9%.
5. Totals hide time trends: machine warming, operator fatigue or humidity drift across the session would otherwise load systematically onto whichever factor happened to run late, biasing its effect. Randomisation converts such drift into noise shared by all cells, which the replicate error then honestly represents.

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